

BORDER OPTICS — Executive Summary

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A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)
Project Overview
The Question
The Method
The Finding
Validation & Robustness Checklist
Honest Limitation
Real-World Relevance

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A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)

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Project Overview

The origins of BORDER OPTICS lie in one line in a Parliamentary reply. Responding directly to the question, the Ministry of Home Affairs said that no impact assessment has ever been conducted for the Vibrant Villages Programme — this despite the scheme having already sanctioned roughly Rs. 4,800 crore across 2,967 villages, covering five Himalayan border states and union territories from Arunachal Pradesh to Ladakh. There's an intersection of three fields that I don't want to consider separately — the gap that I'm digging to fill. It starts with political science, given that VVP is a textbook example of infrastructure that does double duty as both civilian welfare and geopolitical signaling. The next is economics, a large-scale public investment that has never been costed against the results it delivers. And the third is geospatial science, because satellite imagery lets the Government's implicit claim of development be directly tested instead of taken on faith. So, I tested it — 258 individually geocoded villages benchmarked against a district-restricted non-VVP control group of 732 villages in the same districts, using two equally defensible ways of compositing the same satellite record, checked against each other, against a control group, across a third year, across three buffer sizes, against two entirely independent satellite proxies (radar backscatter and a machine-learned built-up classifier), and against an independent, non-satellite ground truth of actual building counts and three confirmed real construction sites. Two earlier versions of this project each reported a headline finding that turned out, on closer inspection, to be a data artifact rather than a real effect — and both times the right call was to fix the artifact and report what was actually left, not to keep the result. What's left, on the study's primary NDBI measure, checked every way this project could check it — including a small-cluster-robust wild bootstrap, a genuine pre-treatment placebo test, and now two cloud-masking conventions once a bug in the second one was found and fixed — is a

clean, and now heavily stress-tested, null. One further check this round complicated that null rather than confirming it. Triangulating the same design against two other satellite proxies found that one of them, a machine-learned built-up classifier that independently validates far more strongly against ground-truth building counts than NDBI does, shows a significant positive effect in both compositing windows, and at three villages with confirmed real construction, that same classifier (and night-lights) detected the increase while NDBI did not. A second thing that initially looked like a second complication — cross-checking the cloud-masking method itself seemed to flip the primary NDBI result from null to significant — turned out under external review to be a bug in that check (a snow-class misclassification), not a real disagreement; fixed and rerun, both cloud-masking conventions now agree. Separately, a border-proximity pattern that an earlier version of this project had flagged as new and unchecked has since been run through the same battery of robustness checks as everything else and survives all of them — a real, if still correlational, secondary finding, and one that runs in the opposite direction from what a securitization hypothesis would predict.

The Question

When directly asked in a Lok Sabha Unstarred Question (Lok Sabha Question No. 508, 3 February 2026) whether any impact assessment had been done of the Vibrant Villages Programme, the Ministry of Home Affairs responded, “No impact assessment has been carried out.” The question this project set out to answer traces back to that gap: did India’s Rs. 4,800 crore border-village development scheme produce measurable physical development on the ground — and can the answer be trusted, or does it depend on which measurement choice you make?

The Method

258 individually geocoded villages, benchmarked against a district-restricted non-VVP control group of 732 villages in the same 14 districts, were tested across 5 Himalayan border states and UTs. The primary satellite record was Sentinel-2 built-up-area change (NDBI) and VIIRS night-lights, 2021–2025, each composited two independent ways (full-year and summer-matched). The before/after change was addressed by Wilcoxon signed-rank tests, the budget/border-proximity relationships by Spearman correlations, and the comparison with the control group by a district fixed-effects Difference-in-Differences model. Beyond that core design: the 2021/2025 comparison was expanded to a 3-point 2021/2023/2025 trend; the fixed 500m extraction buffer was compared against 250m and 1km, pulled the same day to remove any archive-timing confound; the district-FE DiD’s reliance on only 14 clusters was independently checked with an exact wild cluster bootstrap (16,384 sign-flip combinations, not an asymptotic approximation); the border-proximity correlation was stress-tested with leave-one-district-out refits, randomization inference, and a spatial-autocorrelation-aware permutation test; a genuine pre-treatment (2019) baseline was pulled for both treated and control villages so a real parallel-pre-trends placebo test could be run, not just a same-year level check; and the primary NDBI/lights design was triangulated against two fully independent satellite proxies — Sentinel-1 SAR radar backscatter (immune to the cloud-masking concerns optical imagery has) and Dynamic World, a machine-learned per-pixel built-up probability — plus an independent, non-satellite ground truth of Google Open Buildings footprint counts and three specific villages with confirmed real construction.

The Finding

On the primary NDBI/lights design, the result is unchanged from the last complete re-extraction and is now considerably more heavily tested: no significant control-group DiD for either outcome, in either window, at any of the three buffer radii, under the standard cluster-robust specification, under leave-one-district-out, under randomization inference, or under an exact wild cluster bootstrap that does not depend on having enough clusters (only 14 here) for the usual asymptotics to be trustworthy. A genuine 2019-to-2021 placebo test — comparing treated and control villages over a period during which VVP-I could not possibly have had an effect, since it wasn't sanctioned until February 2023 — came back clean in three of its four outcome/window combinations, with the fourth (summer NDBI) borderline and reported as such rather than rounded up to a pass.

Two things did change, and neither is a restatement of the old null. First, the border-proximity correlation an earlier version of this project flagged as new and unchecked has now been run through the same robustness gauntlet as every other result here, and it survives: significant in all 14 leave-one-district-out refits for both the summer-NDBI ($\rho=0.291$) and full-year-lights ($\rho=-0.252$) versions, significant under randomization inference ($p\approx 0.0005$ for both), and — the more conservative test, since these 251 villages are not spatially independent observations — still significant once a spatial-autocorrelation-calibrated permutation null is used instead of treating them as independent ($p=0.0295$ summer-NDBI, $p=0.0035$ full-year-lights). The direction is the opposite of what a securitization hypothesis predicts: built-up and lights change generally increased with distance *from* the border, not toward it, among the villages that did change. This is now a real secondary finding, not an open loose end — though it remains a correlation, not a causal claim, and rests on the same 251-village geometry as everything else in this study.

Second, triangulating the core DiD design against two independent proxies did not simply confirm the NDBI null. SAR backscatter (both polarizations) agrees with NDBI — null in both windows. Dynamic World's built-up probability does not: its control-group DiD is significant in both windows (full-year coefficient $+0.0031$, $p=0.0117$; summer $+0.0074$, $p=0.0013$). That result is harder to dismiss than a stray significant test among many, because an independent validation against actual Google Open Buildings footprint counts shows Dynamic World tracks real building counts far more closely (Spearman $\rho=0.808$) than either NDBI ($\rho=0.399$) or night-lights ($\rho=0.478$) do — and at the three specific villages this project separately confirmed underwent real construction (a hostel at Kaho, a border-terminal building at Walong, solar streetlighting at Musai), Dynamic World and night-lights both registered a genuine before-to-after increase at all three, while NDBI moved in the wrong direction at all three. Whether that means VVP-I produced a small, real, geographically or seasonally specific effect that the primary 500m-buffer NDBI measure is simply not sensitive enough to pick up, or whether Dynamic World's DiD is itself a false positive from a different source of noise, is not resolved here. A village-level cross-check found the three proxies agreeing on the *direction* of change in only 32–63% of individual villages, depending on the pair and window — close to chance — which sharpens this disagreement rather than settling it.

Third: a cloud-masking cross-check that initially looked like a second open sensitivity turned out, on closer inspection, to be a bug. Cross-checking the standard QA60 bitmask against an alternative, stricter, per-pixel Scene Classification Layer (SCL) mask — holding the village sample exactly fixed at the 200 villages both masks could produce a valid composite for, in the first pass — flipped the summer NDBI conclusion from null (QA60, $p=0.449$) to significant and positive (SCL, $p=0.0000045$). An external review of this repository caught the

reason: the SCL mask's "clear" class list included class 11, which is snow/ice, not clear ground — a real risk for June–September composites over Himalayan villages. The mask was fixed to exclude snow and rerun against live Earth Engine. Coverage rose to 251/258 villages (matching QA60's own coverage, no longer skewed toward Arunachal Pradesh), and the SCL result is no longer significant (mean change +0.00112, $p=0.288$) — agreeing with QA60's own null on the identical 251 villages ($p=0.436$). Village-level agreement between the two masks also rose sharply (Spearman $\rho=0.734$, up from the pre-fix $\rho=0.544$). This is reported as a bug this project's own external review caught and this project then fixed and re-verified, not as a second unresolved sensitivity.

Test	Full-Year Window	Summer-Matched Window
NDBI change (Wilcoxon, control-group DiD)	$p = 0.310$ — not significant	$p = 0.473$ — not significant
NDBI DiD, exact wild cluster bootstrap (14 clusters, 16,384 draws)	$p = 0.349$ — not significant	$p = 0.495$ — not significant
Night-lights DiD, exact wild cluster bootstrap	$p = 0.324$ — not significant	$p = 0.465$ — not significant
Buffer-radius sweep (250m/500m/1km), same-day pull	all three null, $n=251$ at every radius	all three null, $n=251$ at every radius
Pre-treatment (2019→2021) placebo DiD	NDBI $p=0.864$, lights $p=0.755$ — clean	NDBI $p=0.064$ — borderline, flagged; lights $p=0.968$ — clean
Border-proximity correlation (H3), spatially corrected	lights: naive $p=5.5 \times 10^{-5} \rightarrow$ spatially-corrected $p=0.0035$	NDBI: naive $p=2.6 \times 10^{-6} \rightarrow$ spatially-corrected $p=0.0295$
SAR backscatter, control-group DiD (triangulation)	VV $p=0.791$, VH $p=0.774$ — null, agrees with NDBI	VV $p=0.717$, VH $p=0.953$ — null, agrees with NDBI
Dynamic World "built," control-group DiD (triangulation)	coef +0.0031, $p=0.0117$ — significant, disagrees with NDBI	coef +0.0074, $p=0.0013$ — significant, disagrees with NDBI
Ground-truth check, 3 confirmed-construction villages	NDBI: wrong direction at all 3 · Lights & Dynamic World: correct direction at all 3	—
Cloud-mask cross-check (QA60 vs. SCL, corrected), same 251 villages, summer NDBI	—	QA60: null ($p=0.436$) · SCL: null ($p=0.288$) — agree, once a snow-masking bug was fixed

Validation & Robustness Checklist

- ✓ Dual compositing-window test (full-year + summer-matched) — NDBI/lights control-group DiD null and in agreement in both
- ✓ Two independent metrics tested separately (NDBI + VIIRS), plus two further independent proxies added for triangulation (SAR backscatter, Dynamic World built-up probability)

- ✓ District-restricted non-VVP control group (732 villages, same 14 districts, DiD design), independently verified free of treated-village contamination
- ✓ Control-group DiD checked further: leave-one-district-out, randomization inference, and an exact wild cluster bootstrap (16,384 sign-flip combinations) that does not depend on having enough of the study's 14 district clusters for standard asymptotics — null on every specification
- ✓ Three-point multi-year trend (2021/2023/2025) — consistent with the two-point null
- ✓ Buffer-radius sensitivity swept (250m/500m/1km), all three pulled the same day — null and in agreement at every radius
- ✓ Cross-checked against sanctioned budget (Parliamentary record) across all three core states
- ✓ Genuine parallel-pre-trends placebo test (a second, 2019 pre-treatment year, both groups, both windows) — 3 of 4 combinations clean, the fourth (summer NDBI) borderline and reported as borderline, not rounded up
- ✓ Border-proximity correlation (H3), once flagged as new and unchecked, now run through leave-one-district-out, randomization inference, and a spatial-autocorrelation-corrected permutation test — survives all three in both directions it was found
- △ Triangulation against two independent proxies did not confirm the NDBI null across the board: Dynamic World's built-up classifier shows a significant control-group effect in both windows, validates more strongly against real building-footprint counts than NDBI does, and correctly detected all three of this study's confirmed real-construction case studies where NDBI did not — an open tension between proxies, not resolved in either direction
- ✓ SCL (Scene Classification Layer) cloud-masking cross-check, run against the QA60 bitmask used throughout — first pass had a bug (snow pixels counted as "clear"), caught by external review; fixed and rerun, and holding the village sample fixed, the two masks now agree: both null
- ✓ requirements-lock.txt (exact pinned dependency freeze) and the git commit hash backing this release have been generated on the researcher's own machine and committed alongside this write-up

Honest Limitation

This project's own first two reported headline results did not hold up once the underlying extraction was checked and fixed — that history is not glossed over here. What's changed in this round is not a third reversal but a genuine complication. The primary design (NDBI and night-lights, control-group DiD) held up under every additional stress test thrown at it this round, including tests specifically chosen because the study's weakest structural feature — only 14 district clusters, below the 30-40+ usually recommended for cluster-robust standard errors to be fully trustworthy — could plausibly have made the null unreliable rather than real; it did not. A genuine pre-treatment placebo test, something the original control-group design could not support with only one pre-period, mostly confirms rather than undercuts that null, with one borderline exception (summer NDBI, $p=0.064$) disclosed rather than smoothed over.

The complication is triangulation. Adding two independent satellite proxies to check the NDBI/lights result was meant to be a robustness check in the ordinary sense — and for SAR backscatter, that's exactly what it was: null, agreeing with NDBI. Dynamic World's built-up classifier did not agree, and the reasons to take that disagreement seriously rather than dismiss it as one significant result among many are not trivial: it is the proxy that independently validates most strongly against real, non-satellite building-footprint counts ($\rho=0.808$ vs. NDBI's $\rho=0.399$), and it is also the proxy that correctly detected all three of this study's own confirmed ground-truth construction

cases, while NDBI moved in the wrong direction at all three. That does not mean VVP-I produced a real, missed effect — night-lights also correctly detected all three ground-truth cases, and yet night-lights' own control-group DiD is null in both windows, so “detects the case-study villages correctly” and “produces a significant aggregate DiD” don't automatically travel together. It means this project cannot currently rule out either explanation: that the primary NDBI measure under-detects a real but modest VVP-I effect at 500m-buffer resolution, or that Dynamic World's significant DiD is itself a different kind of false positive. A village-level check found the three proxies agreeing on direction of change in only about a third to two-thirds of individual villages depending on the pair — near chance — which is evidence the disagreement is real and structural, not a fluke of one aggregate test. This is reported as an open question, the single most important one this round of work raised, rather than resolved in favor of whichever answer would make for a cleaner headline.

A second thing surfaced only while finishing this round's documentation, and it is the third instance of a pattern this project has now seen three times: a result that looked like a real finding turned out, on closer inspection, to be a data or coding problem, not a genuine effect. Cross-checking the QA60 cloud mask this study has used throughout against an alternative SCL-band mask, on the identical 200 villages both masks could then produce a valid summer composite for, flipped the NDBI conclusion from null to significant and positive. Unlike the two earlier reversals, this one was caught by an external review of the repository rather than this project's own internal checks — the SCL mask's “clear” class list was letting snow pixels through as clear ground, which matters for June-September composites in the Himalayas. Fixed and rerun against live Earth Engine, the SCL-valid sample grew to 251 villages (matching QA60's own coverage almost exactly) and the significant result disappeared: SCL is now null ($p=0.288$), agreeing with QA60's own null on the same villages ($p=0.436$). One loose end from the rerun is flagged rather than explained away: coverage grew from 200 to 251 villages after a *stricter* class list was applied, which is the opposite of what should mechanically happen; the likely explanation, given this project's own history with Sentinel-2 archive backfill, is that more scenes became available between the two extractions, independent of the class-list fix — not confirmed, and recorded as open.

The border-proximity correlation is the one place this round's story is a clean upgrade rather than a complication: what was an unchecked, freshly-emerged pattern is now a robustly surviving one, significant across every leave-one-out refit, randomization test, and spatial-autocorrelation correction this project could throw at it — while remaining, honestly, a correlation rather than a causal claim, and one running counter to what a securitization account of VVP-I would predict.

The cloud-masking cross-check turned out to be the third time this project has found and fixed a coding or extraction bug behind what first looked like a real result — after the missing Sikkim imagery and the contaminated control group. That it took an outside reviewer to catch this one, not this project's own test suite, is disclosed here rather than glossed over: this project's own checks are not infallible, and a fresh, external read of the codebase caught something months of internal iteration had not.

The data gaps disclosed in earlier versions of this project — the “unresolved” villages in Ladakh, the “partial” sample in HP, and the “block” confusion in Uttarakhand — are unchanged and still disclosed here. This study's core sample covers Arunachal Pradesh, Sikkim, and Uttarakhand — the three VVP-I states with publicly available, geocoded village-wise lists. Himachal Pradesh is included illustratively only (7 villages) and Ladakh is excluded entirely, because neither jurisdiction has published a village-wise VVP-I list (confirmed against

PIB releases, MHA's own parliamentary replies, and Ladakh's district administration, all of which give only aggregate project counts and budgets). Obtaining one would require an RTI application, which was drafted (`docs/RTI_draft_HP_Ladakh_village_lists.md`, kept for transparency) but not filed, given the cost and multi-month reply timeline relative to this project's scope — a disclosed scope boundary, not an open task.

Real-World Relevance

Parliament was told no independent impact study exists for a Rs. 4,800 crore government scheme spanning 2,967 sanctioned villages — this project is that study, including the parts where its own earlier answers had to be corrected and, this round, complicated rather than simply confirmed. It demonstrates the feasibility of satellite methods for verifying public infrastructure projects, and it demonstrates something else that matters for how this kind of study should be read: a result that survives a battery of robustness checks on its own terms — a control group, a buffer-radius sweep, a small-cluster-robust bootstrap, a genuine pre-treatment placebo test — can still be only part of the picture if the method chosen to measure the outcome is not the only reasonable one available, and triangulating against an independent proxy validated by real ground truth is what surfaced that here. The useful answer for a policymaker isn't "no effect, case closed" or "an effect exists, case closed" — it's that a single remote-sensing index, however extensively cross-checked against everything except a different index, is not automatically the last word, and a study honest about that limitation is more useful than one that picks whichever proxy tells the tidier story.

GitHub: github.com/sakshimaske303-commits/BORDER_OPTICS | Live Dashboard: borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app | Zenodo DOI: [10.5281/zenodo.21759970](https://doi.org/10.5281/zenodo.21759970)

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